Marine Environmental Monitoring & Coastal Ecosystem-Preservation
Literature Review

Paper 1: Towards the Spectral Mapping of Plastic Debris on Beaches

Jenna A. Guffogg, Mariela Soto-Berelov, Simon D. Jones, Chris J. Bellman, Jennifer L. Lavers, Andrew K. Skidmore
Remote Sensing, 2021

Summary

The authors walked two beaches on the remote Cocos (Keeling) Islands with a handheld spectroradiometer and measured how sunlight reflects off patches of real, naturally-washed-up plastic (0.08%–7.94% of the ground covered). They checked six wavelengths already known from lab studies to react to plastic. Only two of them (1215 nm and 1732 nm) actually showed up reliably against sand and seaweed, and even then only 4 of 18 plastic-covered plots looked statistically different from clean sand. So detecting plastic this way works sometimes, but there's no neat rule linking 'more plastic = stronger signal.'

Objectives

- See if known plastic spectral features still work on messy, real beaches (not lab samples).
- Check whether the strength of these features scales with how much plastic covers the ground.
- Build a small field dataset of real beach-plastic spectra to help future satellite detection work.

Research Gap / Limitations

- Almost no prior studies measured real, in-situ beach plastic - most used clean lab samples.
- Sand, moisture and mixed organic debris can hide or mimic the plastic signal.
- No consistent link found between % surface cover and signal strength - only 4 of 18 plots were statistically distinct from bare sand.
- Three of the six candidate wavelengths (931, 1045, 2046 nm) didn't work at all in the field.
- Only one small field trip, two beaches, 20 plots - not much data to generalise from.
- Plastic colour/type (hard vs. translucent) likely affects detection but wasn't tested directly.

Methodology (Tools, Techniques & How They Were Applied)

- ASD FieldSpec Pro 4 spectroradiometer (350–2500 nm) with a white reference panel; QGIS and Agisoft Metashape for mapping plastic cover in each plot; MATLAB for spectral processing.
- 1 m² quadrats surveyed, photographed and digitised to get % plastic cover; spectra smoothed and checked with continuum removal to isolate absorption-feature depth, width and area.
- Linear regression tested cover vs. feature strength; Mann–Whitney tests compared each plastic plot against a sand-only control.

Uniqueness / Novelty

- One of the first studies to spectrally measure plastic exactly as it sits on a real beach, mixed with sand and organic debris.
- Tests realistic, low plastic-cover levels (under 8%) instead of the 20–100% used in earlier lab work.

Paper 2: Influence of Near-Surface Currents on the Global Dispersal of Marine Microplastic

David Wichmann, Philippe Delandmeter, Erik van Sebille
Journal of Geophysical Research: Oceans, 2019

Summary

Most ocean-plastic models only track particles floating right at the surface. This study asks: what if plastic sinks a little (through biofouling or wind mixing) and drifts in currents just below the surface instead? Using a 10-year computer simulation of about a million virtual particles, the authors show that the famous 'garbage patches' get weaker and eventually vanish by about 60 m depth, and that currents just below the surface can carry plastic from the subtropics all the way to the Arctic and Southern Ocean - offering a plausible explanation for why plastic keeps turning up in such remote places.

Objectives

- Test how much currents just below the surface (not just at the surface) matter for where global plastic ends up.
- Compare fixed-depth, randomly-mixed, wind-mixed and full 3D particle transport in the same model.
- Suggest a physical explanation for plastic found in polar regions far from any gyre.

Research Gap / Limitations

- No earlier study had isolated the effect of subsurface currents on global plastic dispersal, despite biofouling being known to sink plastic.
- Surface-only models can't explain why so little of the plastic entering the ocean is ever found floating, or why plastic shows up at the poles.
- The depth scenarios are simplified - real plastic likely bobs up and down through repeated fouling/defouling cycles, which isn't modelled here.
- No degradation, fragmentation or realistic (continuous) plastic input is included; particles are released just once.
- 3D results are sensitive to the arbitrary starting depth (1.5 m) chosen for particles.
- A single, constant mixing/diffusion value is used everywhere, even though real ocean turbulence varies a lot by region.

Methodology (Tools, Techniques & How They Were Applied)

- Parcels Lagrangian particle-tracking software, driven by 10 years of NEMO ocean/wind model data at 1/12° resolution.
- Four transport scenarios run in parallel: particles fixed at set depths (0–120 m), randomly mixed within that range, wind-driven exponential mixing, and full 3D currents.
- About a million particles released once and tracked for 10 years; final positions mapped and compared, and a 'transport matrix' built to see how much each ocean region leaks particles to its neighbours and the poles.

Uniqueness / Novelty

- First study to specifically isolate what subsurface (as opposed to surface) currents do to global plastic transport.
- Offers a new, testable mechanism - subsurface current transport - for how plastic reaches the Arctic and Antarctic.

Paper 3: The Physical Oceanography of the Transport of Floating Marine Debris

Erik van Sebille et al.
Environmental Research Letters, 2020

Summary

This is a big-picture review, written by roughly 40 oceanographers, pulling together everything currently known about the physics that pushes floating plastic around the ocean - from huge gyre currents and eddies down to wind-driven Langmuir stripes and individual breaking waves on a beach. It also reviews how we study all this (drifting buoys, lab tanks, satellites, computer models) and closes with a long list of what we still don't understand.

Objectives

- Pull together, in one place, everything known about the physical processes moving floating plastic at every scale, from ocean gyres to the surf zone.
- Flag the biggest open questions still limiting our understanding.
- Review the tools available (field observation, lab work, remote sensing, modelling) and borrow lessons from related fields like oil-spill tracking.

Research Gap / Limitations

- There's a huge mismatch between how much plastic is thought to enter the ocean each year and how little is ever found floating - the 'missing plastic' problem is still unsolved.
- Vertical mixing processes (Langmuir turbulence, wave breaking, fronts) are poorly represented in most models.
- It's largely unknown which sinking mechanism (marine snow, fecal pellets, biofouling, etc.) matters most, or where.
- Coastal and beaching models are far less developed than open-ocean models; stranding and resuspension are barely understood.
- Almost nothing is known about how storms, floods or tsunamis affect debris transport.
- Satellites still can't see submerged or micro-sized plastic, and telling plastic apart from foam, sunglint or clouds remains genuinely hard.

Methodology (Tools, Techniques & How They Were Applied)

- A structured literature synthesis by an international working group (SCOR WG153), built from workshop discussions, targeted searches and contributions from experts worldwide.
- Draws on drifter/buoy data, laboratory flume experiments, satellite remote sensing (optical, thermal, radar) and both Eulerian and Lagrangian ocean-model frameworks.
- Organised by spatial scale - open ocean, mesoscale eddies, Stokes drift, Langmuir circulation, sea ice, river plumes, coastal/surf zone - with each process's typical strength and depth range summarised in tables and diagrams.

Uniqueness / Novelty

- First review to focus purely on the physics of floating debris transport, across the full range of scales in one document.
- Deliberately borrows working knowledge from oil-spill, search-and-rescue and plankton-transport research and applies it to plastic.

Paper 4: MARIDA: A Benchmark for Marine Debris Detection from Sentinel-2 Remote Sensing Data

Katerina Kikaki, Ioannis Kakogeorgiou, Paraskevi Mikeli, Dionysios E. Raitsos, Konstantinos Karantzalos
PLOS ONE, 2022

Summary

MARIDA is an open dataset built to train computers to spot marine plastic in free Sentinel-2 satellite images - and, importantly, to tell it apart from things that look similar from space, like seaweed, ships, foam, waves and cloudy or turbid water. It's made of about 1,400 image patches from 63 satellite scenes across 11 countries, hand-labelled by three experts. The authors also test a few standard machine-learning models (Random Forest and U-Net) on it, getting a decent but imperfect accuracy - proof the dataset is a genuinely useful, if still challenging, benchmark.

Objectives

- Build a realistic, open, worldwide dataset of Sentinel-2 imagery for training marine-debris detection models.
- Make sure the dataset separates real debris from 14 other look-alike sea/sky features, not just debris-vs-water.
- Provide and test baseline machine-learning models so future researchers have something to compare against.

Research Gap / Limitations

- Earlier debris datasets were small, geographically narrow, or focused on just one object type (only ships, only clouds, etc.).
- Plastic is genuinely hard to tell apart from waves, sunglint, foam and clouds because it varies so much in colour and how much of it sits underwater.
- The dataset itself isn't evenly spread around the globe, simply because there isn't enough open in-situ litter-reporting data everywhere.
- The 'Natural Organic Material' class (driftwood, vegetation) is spectrally almost identical to plastic debris, so every model tested struggled with it.
- Some known debris events had to be dropped because of cloud cover or bad atmospheric correction.
- Labels are sparse rather than 'pure' pixels, and boundary cases (like a moving ship's wake) were genuinely hard even for expert annotators.

Methodology (Tools, Techniques & How They Were Applied)

- Sentinel-2 imagery, ACOLITE atmospheric correction, and two custom spectral indices (FDI, FAI) to help spot floating material.
- Three experts independently labelled every image at pixel level with a confidence score; disagreements were resolved with a formal agreement protocol, and t-SNE was used to catch and fix mislabelled outliers.
- Random Forest (three versions, adding spectral indices and texture features) and a U-Net neural network were trained and tested on the finished dataset, scored with IoU, F1 and pixel accuracy.

Uniqueness / Novelty

- First open Sentinel-2 dataset that separates marine debris from 14 realistic look-alike classes, not just a simple debris/water split.
- Globally spread, multi-season, deliberately 'messy' (non-ideal) imagery, with both pixel-level and image-level labelling tasks included.

Paper 5: Finding Plastic Patches in Coastal Waters using Optical Satellite Data

Lauren Biermann, Daniel Clewley, Victor Martinez-Vicente, Konstantinos Topouzelis
Scientific Reports, 2020

Summary

This paper introduces a new index - the Floating Debris Index (FDI) - for spotting patches of floating plastic in free Sentinel-2 satellite images, then uses it to build a small 'fingerprint library' of what plastic, seaweed, foam, timber and pumice each look like spectrally. Feeding those fingerprints into a simple Naïve Bayes classifier, the authors correctly sorted suspected plastic from other floating stuff about 86% of the time across four very different coastlines (Ghana, Vietnam, Canada and Scotland) - and got 100% right on plastic targets they knew were really there.

Objectives

- Check whether floating macroplastic patches can actually be seen in free Sentinel-2 satellite data.
- Test whether plastic can be automatically told apart from natural floating material like seaweed and foam.
- Build and validate a new spectral index (FDI) designed specifically for Sentinel-2's bands.

Research Gap / Limitations

- Very few earlier studies had managed to detect floating plastic from satellite at all - older satellites simply didn't have the right resolution.
- A single stray bottle or bag is too small to see from space; only larger, wind- or current-aggregated patches are detectable.
- Only two solid sources of validated plastic data existed to train on (deployed targets in Greece, and a flood event in Durban), which is a small sample.
- Atmospheric correction introduces its own uncertainty, and different correction methods give noticeably different results.
- The method works best in fairly clear water and hasn't been tested much in turbid, complex coastal areas.
- The four case-study regions only had 'suspected' plastic, without on-the-ground confirmation like the training sites had.

Methodology (Tools, Techniques & How They Were Applied)

- Sentinel-2 imagery, ACOLITE atmospheric correction, and the newly built Floating Debris Index (a modified Floating Algae Index using the Red Edge band as a baseline).
- Spectral signatures built for plastic, seaweed, timber, foam and pumice from a mix of deliberately deployed plastic targets and a real, uncontrolled pollution event.
- A Gaussian Naïve Bayes classifier, trained on FDI, NDVI and a few reflectance bands, sorted detected pixels into plastic vs. other material, checked against a confusion matrix.

Uniqueness / Novelty

- First study to show floating plastic can be reliably told apart from lookalike natural material using only free Sentinel-2 data, across several real coastlines.
- Introduced the FDI index, which has since become a standard tool in later marine-debris studies (including MARIDA).